Samara Luanne Pires

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Data Science & AI Professional | Machine Learning, Statistical Modelling & Business Analytics

Combining hands-on technical expertise with strong analytical thinking and business understanding, I specialize in turning complex problems into practical, data-driven solutions. I bring experience in developing AI systems, applying statistical and machine learning techniques, and translating business requirements into technical solutions, with a focus on building reliable systems that deliver measurable value.

Professional Experience

AltaML - Associate Business Solution Consultant: *May 2026 - Aug 2026*

- Translated Government of Alberta registry workflows into product requirements, acceptance criteria, and technical specifications for an LLM-powered research assistant, aligning stakeholder needs with AI engineering solutions.
- Developed and refined agentic RAG workflows using LLMs, prompt engineering, query rewriting, and retrieval techniques, contributing to AI agent functionality, response relevance, and escalation handling.
- Designed and executed evaluation scenarios to assess retrieval accuracy, response completeness, and AI agent reliability, identifying failure modes and recommending improvements to system performance.

Amii (Alberta Machine Intelligence Institute) - Student Trainee: *Sep 2025 - Nov 2026*

- Investigated LLM use cases and agentic-workflow design with 4 AI-focused startups, applying prompt engineering and analytical problem-solving to early-stage product challenges.

Data For Good Edmonton - Data Analyst: *Sep 2025 - Apr 2026*

- Built Python/NumPy scenario models combining cyclist counts, epidemiological risk factors, and healthcare cost assumptions to estimate 10-year health and economic impacts of protected cycling routes.

LTIMindtree - Quality Analyst - ***Client: Johnson Controls*** *: Jun 2021 - Oct 2024*

- Developed Python/Selenium testing automation for enterprise applications, reducing manual testing effort by 81% and execution time by 40%.
- Partnered with cross-functional teams to translate business requirements into validation plans for a UK/Ireland customer portal, supporting delivery and a 5/5 client satisfaction rating.

Expenzing - Data Analyst Intern: *Dec 2019 - Jan 2020*

- Analyzed expense and payment data and built Power BI dashboards that supported 23% cost savings and a 32% improvement in reporting efficiency.

Education

University of Alberta - Master of Science in Modelling, Data & Predictions | 2025 - Dec 2026 | GPA - 3.7 / 4
Relevant Focus: Statistical modeling, time series analysis, regression methods, and machine learning for prediction and decision-making
University of Mumbai, Don Bosco Institute of Technology - Bachelor of Engineering, Computer Science | 2017 - 2021
Relevant Coursework: Introduction to Machine Learning; Statistical Analysis; Data Visualization; Python Programming

Skills & Certifications

Programming & Data: Python, R, SQL, PostgreSQL, MATLAB, Pandas, NumPy, Matplotlib
Statistics & Machine Learning: Statistical Inference, Hypothesis Testing, Exploratory Data Analysis (EDA), Logistic Regression, Bayesian Modelling, Predictive Modelling, Forecasting
AI & LLMs: AI Agent Development, Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), Prompt Engineering, Query Rewriting, Agentic Workflows, LLM Evaluation
Business Intelligence & Visualization: Power BI, Tableau, Excel, Power Query, Power Pivot, DAX
Developer & Testing Tools: Git, GitHub, Jupyter Notebook, Selenium, BrowserStack
Soft Skills: Communication, Stakeholder Engagement, Problem-Solving, Cross-Functional Collaboration, Data Storytelling, Presentation Skills
Certifications: SQL for Data Science (UC Davis) , ISTQB Foundation(GASQ), Google Gen AI Leader, Google Professional Data Engineer
Awards: Runner Up - Best Final Year Thesis, 2026 - Data Masters Challenge - 2nd Place

Projects

Public-Sector Agentic RAG - MSc Capstone | Python, LLMs, RAG, Prompt Engineering

Developed document ingestion and validation workflows for 20 Alberta government sources, establishing a verified knowledge base for a public-sector AI research assistant.

Designed retrieval evaluation scenarios and information-quality checks to assess answer accuracy, source reliability, and evidence-grounded responses.

Nexora Commerce - Business Intelligence & Forecasting | Excel, Power Query, Power Pivot, DAX

Analyzed 1.06 million retail transactions using Excel, Power Query, and Power Pivot to identify sales trends, revenue patterns, and business performance insights.

Developed interactive KPI dashboards and forecasting analyses using DAX to support performance monitoring and data-driven business decisions.

Buy Now, Pay Later - Credit Risk Prediction | Python, Logistic Regression, Bayesian Modelling

Analyzed approximately 50,000 financial transactions using Python to explore customer behaviour, identify credit-risk patterns, and prepare datasets for predictive modelling.

Applied logistic regression and Bayesian modelling techniques to investigate relationships between customer characteristics and credit risk.

Research Experience & Publications

Cyber Threat Analysis Using Pearson & Spearman Correlation via EDA - ICSCCC 2023, IEEE

Conducted exploratory data analysis and statistical modeling on the UNSW-NB15 dataset, applying Pearson and Spearman correlation, chi-square tests, and t-tests to analyze feature relationships, identify cyber-attack patterns, and support predictive security modeling.

A Real-Time Position Monitoring System for Fall Detection & Analysis - ICNTE 2021, IEEE

Developed and evaluated machine learning–based fall detection models using OpenPose and SSD-MobileNet, achieving 93% classification accuracy in real-time posture recognition for health monitoring applications.

Fall Stop: Real-Time Fall Detection & Vital Monitoring - INDISCON 2021, IEEE

Engineered an IoT-enabled machine learning system integrating SVM-based classification with real-time sensor data and automated alerts, focusing on model inference, data preprocessing, and response-time optimization.

Extracurricular Activities & Leadership

Co-Founder, DBIT SevaTech (2019–2021): Co-founded a student-led technology initiative delivering digital solutions for nonprofit organizations, coordinating student teams and creating 40+ internship opportunities through industry and community engagement.

STEM Outreach, SIGMAS: Supported Edmonton mathematics outreach, including a fair engaging 900+ students.

Additional: Taekwondo (Blue Belt), swimming, certified professional baker, French (A1–A2)